



ML BASED FRAUD TRANSACTION DETECTION SYSTEM

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This project benefited from open-source technologies and publicly documented machine-learning methods. Their communities made it possible to construct a complete prototype spanning transaction ingestion, feature engineering, model training, model comparison, risk policy configuration, case management, and suspicious transaction report generation.

DECLARATION

I declare that this project report is an original account of the work completed for the Intelligent Fraud Transaction Detection and AML Case Management System. Material derived from external publications and software documentation is acknowledged in the references. The system is an academic prototype and does not claim that an anomaly is legally proven fraud; suspicious outcomes require authorized human review.

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ABSTRACT

Financial institutions process large volumes of transactions while needing to identify unusual activity quickly, explain decisions to analysts, and preserve evidence for regulatory review. A single machine-learning model is insufficient because institutions may have no confirmed fraud labels, transaction behavior changes as history grows, customer-specific patterns differ from peer behavior, and model predictions alone do not represent deterministic anti-money-laundering obligations.

This project implements an Intelligent Fraud Transaction Detection and AML Case Management System using Angular, Spring Boot, FastAPI, and Microsoft SQL Server. A global setting selects unsupervised or supervised learning. Unsupervised mode uses Isolation Forest, Autoencoder, and Local Outlier Factor; supervised mode uses XGBoost, Random Forest, and Logistic Regression. Both modes feed a layered risk score combining customer behavior, peer behavior, the selected ML ensemble, and deterministic AML rules. Weights normalize to 100%, MEDIUM and HIGH outcomes create cases, and HIGH outcomes generate draft STR XML after transaction commit.

The main research feature evaluates how model behavior changes when the oldest 10%, 25%, 50%, and 100% of an immutable snapshot are used. Label-free diagnostics are reported for unsupervised models, while labelled chronological evaluation is used for supervised models. Scalability is addressed through persisted point-in-time features, closed business dates, keyset export, checksummed Parquet snapshots, bounded evaluation, asynchronous case work, pagination, and versioned artifacts. The result separates anomaly evidence from confirmed fraud and supports analyst review, false-positive feedback, manual cases, and STR preparation.

Keywords: fraud detection, anomaly detection, anti-money laundering, supervised learning, unsupervised learning, model comparison, explainable risk scoring, case management, STR XML, scalable architecture

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CHAPTER 1

INTRODUCTION

1.1 Background

Fraud monitoring and anti-money-laundering operations must identify unusual transactions without interrupting legitimate banking activity. Conventional rule engines are transparent but can be rigid and generate large numbers of false alerts. Machine-learning methods can discover complex patterns, but their raw scores are difficult to compare, may drift as the amount of history changes, and can fail silently when feature engineering differs between training and prediction.

The project began as an anomaly detector for a bank transaction dataset containing transaction identity, account, amount, date, type, location, channel, customer age, occupation, login attempts, balance, and previous transaction date. It evolved into a broader AML decision-support system that persists point-in-time features, compares models on growing chronological datasets, supports labelled and unlabeled learning, explains layered risk contributions, and manages the analyst workflow after a suspicious decision.

1.2 Motivation

A bank may store millions of transactions and cannot repeatedly send its entire lifetime history through one synchronous request. It may also lack reliable fraud labels because alerts are not equivalent to confirmed fraud. Even when a model detects an anomaly, analysts need to know whether the amount was unusual for the customer, unusual for comparable peers, associated with velocity or structuring rules, or simply outside a model boundary. These operational and scientific limitations motivated a system that separates evidence layers and preserves an audit trail.

1.3 Problem Statement

- Transaction datasets often have no trustworthy class label, preventing valid accuracy, precision, recall, and F1 claims.

- A model that looks good on a small sample may behave differently when historical data grows.
- Raw scores from different algorithms use incompatible scales and cannot be added directly.
- Scoring one transaction can become slow if the system scans large historical tables or retrains models synchronously.
- Customer behavior, peer behavior, AML rules, and ML evidence can disagree and require a governed method of combination.
- An anomaly alert alone is not a complete operational outcome; investigators need cases, explanations, notes, false-positive handling, and STR preparation.
- Administrative functions require authentication, authorization, model versioning, and reproducible training data.

1.4 Project Objectives

1. Build an end-to-end web platform for upload, transaction checking, training, model comparison, configuration, case review, and STR XML generation.
2. Support unsupervised and supervised learning through one system-level mode while keeping their models and evaluation metrics scientifically separate.
3. Measure how model behavior changes using the oldest 10%, 25%, 50%, and 100% of an immutable dataset.
4. Combine transparent customer behavior, peer behavior, ML evidence, and deterministic AML rules into a normalized 0-100 risk score.
5. Make suspicious decisions explainable through reason codes, component scores, model diagnostics, and persisted evidence.
6. Design training, data export, and real-time decision paths that can evolve toward bank-scale workloads.

1.5 Scope

The implemented scope includes CSV and Excel ingestion, real-time transaction creation, persisted feature vectors, cold-start settings, global learning mode, six supported

model choices across two learning modes, mode-aware tuning, chronological model comparison, weighted risk policy configuration, JWT authentication, role authorization, automatic and manual cases, pagination, analyst notes, false-positive marking, and draft STR XML. The project is an academic prototype: it does not connect to a production core-banking system, submit reports to a regulator, or replace human adjudication.

1.6 Significance

The primary contribution is not a claim that one algorithm proves fraud. The contribution is an integrated framework for selecting and governing evidence: it tells users which methodology is valid for their data, how models behave as history grows, how non-model rules affect the final result, and why a case was generated. This makes the prototype useful for academic evaluation and for discussing production AML design constraints.

CHAPTER 2

BACKGROUND AND RELATED APPROACHES

2.1 Rule-Based Transaction Monitoring

Traditional monitoring uses deterministic thresholds for structuring, velocity, unusual amount, sanctions, location, and channel behaviour. Rules are easy to explain and can implement mandatory controls, but static thresholds may not reflect individual customer baselines. This project therefore retains rules as one explicit layer rather than asking ML models to rediscover every legal or operational control.

2.2 Unsupervised Anomaly Detection

When labels are unavailable, the system uses three complementary detectors. Isolation Forest isolates rare observations through randomized trees; the Autoencoder learns common feature patterns and treats high reconstruction error as unusual; Behavioral Cluster Outlier measures cluster-conditional deviation from comparable transaction behaviour. Their outputs indicate abnormality rather than confirmed fraud.

2.3 Supervised Fraud Classification

When an auditable binary label is available, supervised classifiers learn relationships between point-in-time features and reviewed outcomes. XGBoost provides scalable nonlinear boosting, Class-Balanced Random Forest provides a robust bagged-tree benchmark, and Extra Trees adds stronger split randomization and computational efficiency. Fusion strategies remain research comparisons; the production selector exposes only the three base classifiers. The comparison protocol emphasizes precision-recall measures because fraud is highly imbalanced.

2.4 Explainable Layered Risk

Pure voting creates unintuitive outcomes when three models agree but the ML layer owns only part of the total risk budget. The project replaces this ambiguity with an explicit layered score. Each component produces a normalized value between zero and one; configured top-level weights sum to one; selected ML model weights also sum to one

inside the ensemble. The UI shows both anomaly votes for diagnostics and the weighted score that actually determines the risk level.

2.5 Research Gap Addressed

Many demonstrations train one model on one fixed dataset and report either accuracy without proving label quality or anomaly counts without studying stability. This project adds a chronological growth experiment, separates label-free and label-based metrics, includes transparent non-ML layers, and connects evaluation to a governed operational workflow. It therefore studies both model behaviour and system behaviour.

2.6 Research-Guided Design Decisions

The project follows published methods as design guidance rather than claiming to reproduce every paper exactly. Each source contributes a concrete decision that is visible in code, evaluation, or governance.

Research source	Decision adopted in this system
Fraud Detection Handbook (Le Borgne et al., 2022)	Reproducible simulator data, chronological evaluation, severe class imbalance, PR-oriented metrics, and investigator-capacity awareness.
Liu, Ting and Zhou (Isolation Forest)	A scalable label-free detector that isolates rare observations and supports bounded subsampling.
Sakurada and Yairi (Autoencoder anomaly detection)	Reconstruction error represents deviation from learned normal behaviour.
Goix, Sabourin and Clemencon (Excess-Mass curves)	EM-AUC is used as a label-free quality diagnostic instead of inventing accuracy without labels.
Breiman; Geurts et al.; Chen and Guestrin	Class-Balanced Random Forest, Extra Trees, and XGBoost form the supervised base-model comparison.
Bach Nguyen et al. (future information)	Real-time scoring uses past-only features; later transactions may be used only for posterior investigation or later retraining, preventing online leakage.

CHAPTER 3

SYSTEM ANALYSIS AND ARCHITECTURE

3.1 Stakeholders and Use Cases

Administrators configure the learning mode, tuning values, risk policy, training runs, and model governance. Reviewers inspect cases, search transactions, add notes, mark false positives, and generate STR XML. External transaction sources upload files or call the transaction API. The ML service trains and scores models but does not own business records or case decisions.

Table 1. Functional requirements

Actor	Major functions
Administrator	Settings, uploads, training operations, model tuning, comparison, risk policy, governance
AML Reviewer	Transaction checks, case search, notes, false-positive review, manual case creation, STR XML

Actor	Major functions
Transaction Source	Bulk CSV/Excel upload or real-time transaction API
ML Service	Feature alignment, model training, prediction, comparison metrics, artifact persistence

3.2 Non-Functional Requirements

Table 2. Non-functional requirements

Quality	Requirement and implementation direction
Security	Stateless JWT authentication, BCrypt password hashes, server-side role authorization, deny-by-default API policy
Explainability	Reason codes, component scores, model diagnostics, feature summary, policy and model versions
Scalability	Persisted features, Parquet snapshots, bounded evaluation, asynchronous workflows, pagination
Integrity	Closed business dates, chronological cutoffs, checksums, immutable model bundles, duplicate protection
Availability	Auditable fallback when the ML service is unavailable; case creation occurs after commit
Maintainability	Separate Angular, Spring Boot, and FastAPI applications with stable REST contracts

3.3 Overall Architecture

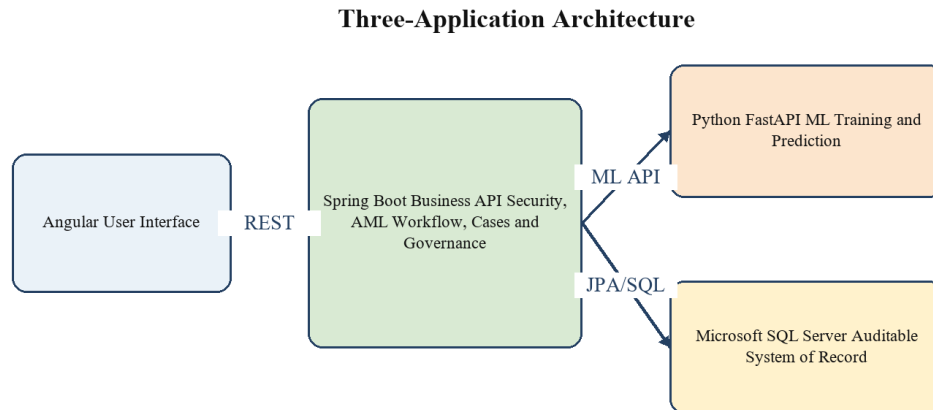


Figure 1. Three-application architecture

Angular provides the browser interface. Spring Boot owns transaction persistence, security, feature orchestration, risk policy, model governance, and case workflow. FastAPI owns numerical preprocessing, model fitting, artifact loading, prediction, and research analysis. SQL Server is the auditable system of record. This division prevents the Python service from directly mutating business records and prevents the UI from accessing model artifacts or database tables.

3.4 Database and Audit Design

Core operational records include transactions, upload batches, persisted transaction features, learning eligibility, prediction logs, fraud alerts, cases, case notes, STR reports, application configuration, training runs, immutable model registry entries, validation reports, deployment pointers, and growth studies. Migration scripts are additive and independently verifiable. Training runs reference closed business windows and immutable exported datasets rather than mutable lifetime queries.

3.5 Transaction Decision Flow

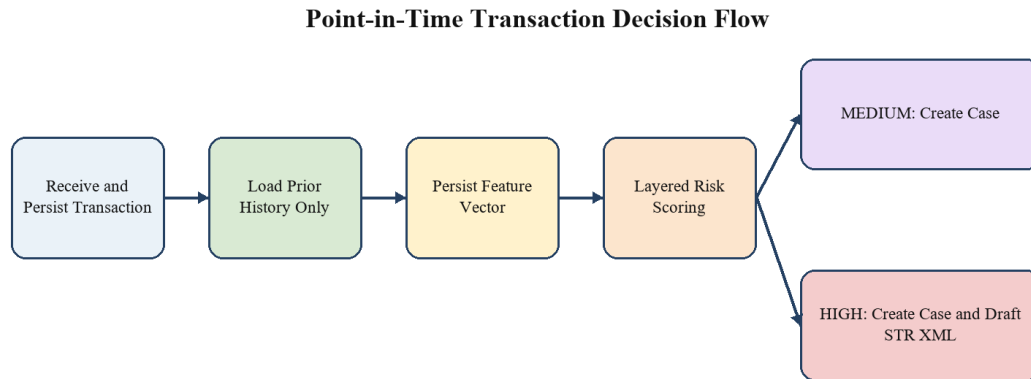


Figure 2. Point-in-time transaction decision flow

A transaction is first persisted with processing status. Feature engineering loads only records earlier than the transaction timestamp, preventing future information leakage. The feature vector is persisted before prediction. The layered orchestrator returns the final risk score, model evidence, feature summary, reasons, and action. Suspicious results create an alert and publish an after-commit event. MEDIUM creates a case; HIGH creates a case and automatically generates draft STR XML asynchronously so the transaction database commit is not held open by case work.

3.6 Authentication and Authorization

Login issues a signed JWT. Passwords are stored using BCrypt with strength 12. Spring Security permits health, login, registration, and API documentation without authentication, requires authentication for all other APIs, and restricts upload, training, AML writes, and comparison writes to ADMIN or AML_ADMIN. Angular route guards improve usability, but the backend remains the authority.

CHAPTER 4

METHODOLOGY

4.1 Global Learning Mode

The Settings page stores `system.learning_mode` in `app_config`. UNSUPERVISED is selected when confirmed labels are absent. SUPERVISED is selected only when an auditable FraudLabel is present. Upload validation, the model catalog, Model Tuning, Training Operations, the comparison page, and the Risk Policy model list read this mode so the workflow remains consistent.

Table 3. Dual learning model catalog

Mode	Models	Required data	Primary interpretation
Unsupervised	Isolation Forest; Autoencoder; Behavioral Cluster Outlier	Transactions without confirmed class labels	Anomaly evidence and stability
Supervised	XGBoost; Class- Balanced Random Forest; Extra Trees	Reviewed binary FraudLabel	Estimated suspicious probability and classification quality

4.2 Feature Engineering and Leakage Prevention

Features describe amount and balance ratios, hour and day, night and weekend indicators, customer history count, rolling amount statistics, z-scores, debit and credit ratios, transaction velocity, unique beneficiaries, new location/channel/device indicators, peer-group statistics, expected turnover deviation, customer type, and risk rating. Each feature is computed using records available before the prediction time. The feature schema and column order are persisted with every model bundle so training and prediction cannot silently disagree.

4.3 Customer Behavior Scoring

The customer layer asks whether the transaction is unusual for the same customer. A transparent weighted formula combines amount deviation, transaction frequency, time-gap deviation, categorical novelty, and unusual hour. A confidence multiplier reduces customer-specific evidence during cold start. This prevents two or three prior transactions from being treated as a mature profile while retaining limited evidence.

4.4 Peer Behavior Scoring

The peer layer compares the transaction with similar customers. Groups are derived from customer attributes such as occupation and age band, not hard-coded for one profession. If a specific occupation-age baseline lacks enough history, the scorer falls back to a broader parent segment and then to a global baseline, with decreasing confidence. The peer window is anchored to the training snapshot cutoff rather than the newest just-arrived transaction, preventing one future transaction from shifting the historical reference window.

4.5 Deterministic AML Rules

Rules evaluate transparent conditions such as amount multiples, customer and peer z-scores, high transaction velocity, repeated near-threshold activity, expected-turnover deviation, unusual locations, and hard compliance overrides. Rule thresholds are configured separately from model tuning. A hard override remains identifiable as a rule decision rather than being misrepresented as model consensus.

4.6 Layered Weighted Risk Aggregation

Layered Risk Scoring (All Weights Normalize to 100%)

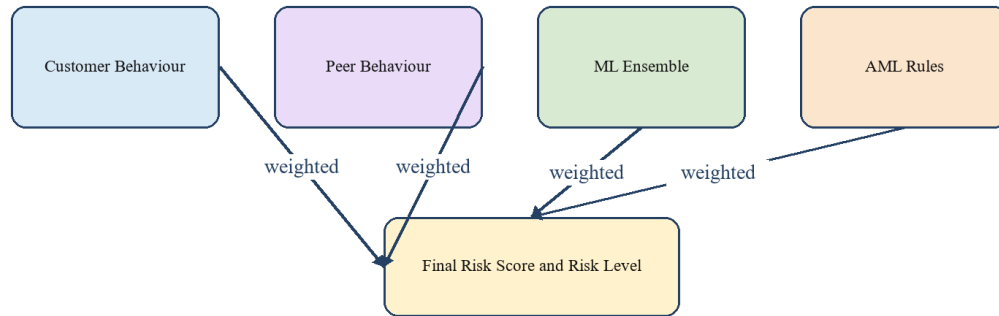


Figure 3. Layered risk scoring

The final normalized score is $\text{customerWeight} \times \text{customerScore} + \text{peerWeight} \times \text{peerScore} + \text{mlWeight} \times \text{mlScore} + \text{rulesWeight} \times \text{rulesScore}$. Top-level weights total 1.0. Inside the ML ensemble, enabled model weights also total 1.0. The UI provides component-level normalization so an operator can increase one value to use the remaining percentage without unpredictably changing unrelated sections. Risk thresholds convert the final 0-1 value to a 0-100 display and NORMAL, LOW, MEDIUM, or HIGH.

4.7 Chronological Growth Comparison

The research experiment orders the immutable snapshot by transaction date and trains each selected model on the oldest 10%, 25%, 50%, and 100% that meet the minimum-row rule. Each partition uses a chronological holdout, so future rows are not randomly mixed into training. This directly answers how behavior, stability, anomaly rate, training time, and throughput change as historical evidence increases.

Table 4. Evaluation metrics by learning mode

Mode	Metrics	What they establish
Unsupervised	EM-AUC, score separation, skewness, anomaly rate, rate control, stability, throughput	Label-free diagnostic quality; not fraud accuracy

Mode	Metrics	What they establish
Supervised	PR-AUC, ROC-AUC, precision, recall, F1, Brier score, confusion matrix	Classification and probability quality on reviewed labels
Both	Training duration, learned rows, evaluation rows, prediction latency	Operational scalability and reproducibility

4.8 Governed Training

Governed Training and Comparison Pipeline

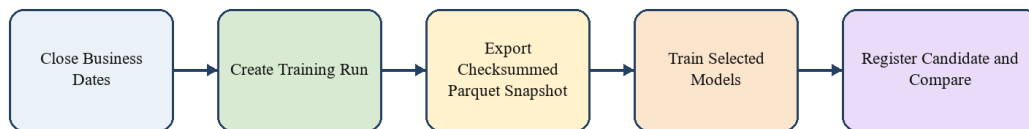


Figure 4. Governed training and comparison pipeline

An administrator chooses when to train; automatic daily and weekly schedules were removed. Business dates must first be closed. Spring Boot exports eligible persisted features through keyset pagination into checksummed Parquet parts. Python trains only the models selected from the active mode and saves versioned artifacts. Spring registers candidates with metrics and metadata. Training does not silently activate a model.

CHAPTER 5

IMPLEMENTATION

5.1 Tools and Technologies

Table 5. Main implementation technologies

Layer	Technology	Purpose
Frontend	Angular 21, TypeScript, RxJS	Responsive pages, forms, route guards, API integration, alerts
Business backend	Java 17, Spring Boot 4, Spring Security, JPA	Transactions, security, policies, cases, training orchestration
ML service	Python, FastAPI, pandas, NumPy, scikit-learn, XGBoost	Training, prediction, evaluation, artifact handling
Database	Microsoft SQL Server	Auditable operational and governance state
Data interchange	REST/JSON, multipart upload, Parquet, XML	Online calls, datasets, and STR draft output

5.2 Angular User Interface

The UI uses a white, responsive dashboard with a toggleable side menu and consistent success/error popup alert. Current routes include Uploads, Datasets and model comparison, Supervised Comparison, Training Operations, Risk Policy, Settings, Model Tuning, Transaction Check, Case Management, and Manual Case Creation. Administrative pages are route-protected, and every page delegates authority to backend APIs.

5.3 Risk Policy Configuration

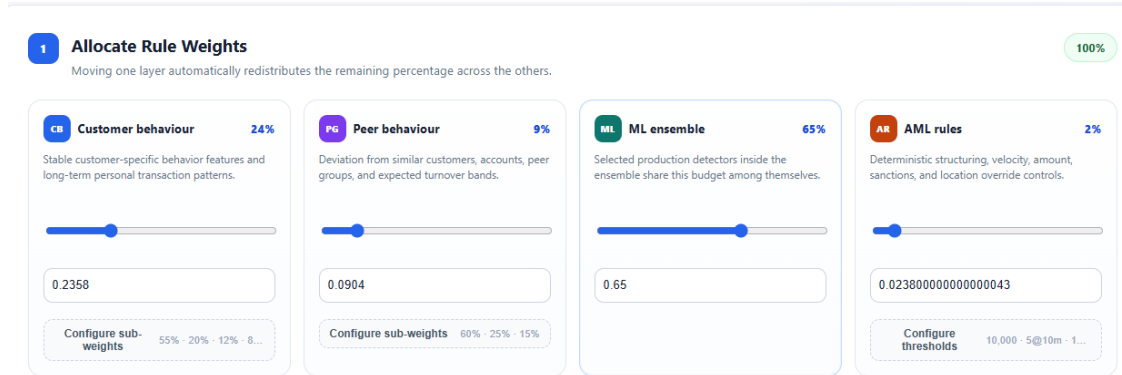


Figure 5. Risk policy configuration interface

The configuration interface allocates 100% across customer behaviour, peer behavior, the ML ensemble, and AML rules. A modal selects active production models based on the global learning mode. Disabled models do not participate. The page displays effective model contribution to the total score and prevents totals above 100%.

5.4 Model Tuning

Model Tuning is mode-aware. In unsupervised mode it exposes controls for Isolation Forest, Autoencoder, and LOF. In supervised mode it exposes XGBoost tree count, depth, learning rate, row and feature sampling; Random Forest tree count, depth, and leaf size; and Logistic Regression regularization and iteration limits. Shared evaluation controls include minimum rows, holdout fraction, maximum evaluation rows, and random seed. Values are persisted in app config and consumed by both training and growth analysis.

5.5 Transaction Check and Explainability

Transaction details

Reset

Transaction ID	Account ID
TXN-1786436891687	AC00455
Amount	Account balance
2000	Optional
Transaction type	Channel
TRANSFER	ATM
Transaction date	Location
08/11/2026 08:28 AM	Mesa
Customer age optional	Customer occupation optional
68	Doctor
Login attempts	
2	

Check Transaction

Figure 6. Transaction entry interface

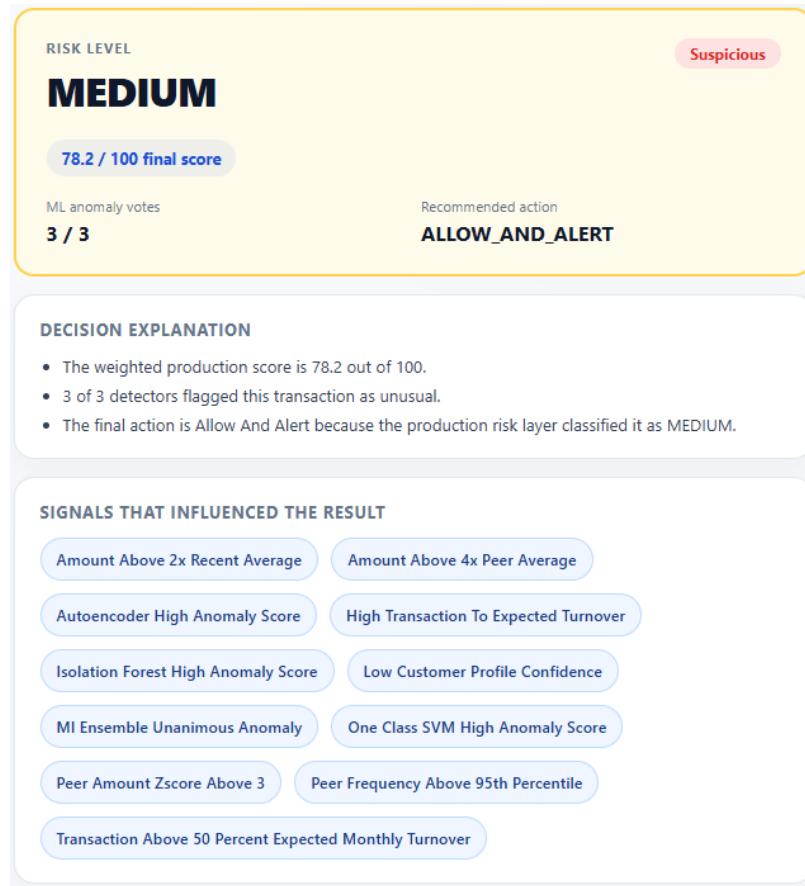


Figure 7. Explainable transaction decision

The transaction page accepts a transaction ID, account, amount, balance, type, channel, date, location, age, occupation, and login attempts. The result displays final score, risk level, action, ML anomaly votes, human-readable reason tags, key features, model diagnostics, latency, layered component scores, policy version, and raw payload for technical review. This separation prevents the raw model decision from being confused with the final policy decision.

5.6 Case Management and STR

Suspicious transactions create fraud alerts and cases after the transaction commit. Case lists use server-side pagination, filtering, and summary counts to avoid loading the full case table. The detail view includes transaction evidence, prediction evidence, analyst notes, reason explanations, and allowed actions. A reviewer can mark a case false positive or generate draft STR XML, but the state machine prevents STR generation from a false-

positive case and prevents an STR-generated case from later being marked false positive. Manual cases reuse existing transactions and avoid duplicates.

5.7 Training and Artifact Management

Training operations expose model checkboxes from the active Risk Policy rather than fixed daily or weekly training types. One run can train selected candidates from the same immutable snapshot. Artifacts include the fitted estimator, scaler or preprocessing state, ordered feature columns, threshold, hyperparameters, metrics, and dataset checksum. Atomic replacement avoids partially written model directories, and model loading caches versioned artifacts for prediction.

5.8 Failure Handling

If FastAPI is unavailable, Spring records the failure and returns the documented safe fallback instead of corrupting transaction state. Invalid roles receive JSON 401 or 403 responses. Duplicate transaction IDs are rejected. Training failures preserve messages and do not activate candidates. Dataset export rejects open business dates. Artifact checksum and feature-version mismatches block governance actions.

CHAPTER 6

5.9 Application Page Walkthrough

This section presents the implemented Angular pages in operational order. Each page is connected to backend controls, persisted evidence, or investigator actions described in the preceding sections.

Interface evidence. Screenshots document the working prototype as tested on 12 August 2026. Numerical results are demonstration evidence from synthetic or label-free datasets and are not claims of bank-production fraud accuracy.

5.9.1 Login and Access Control

The page provides a focused entry point without exposing operational navigation before authentication.

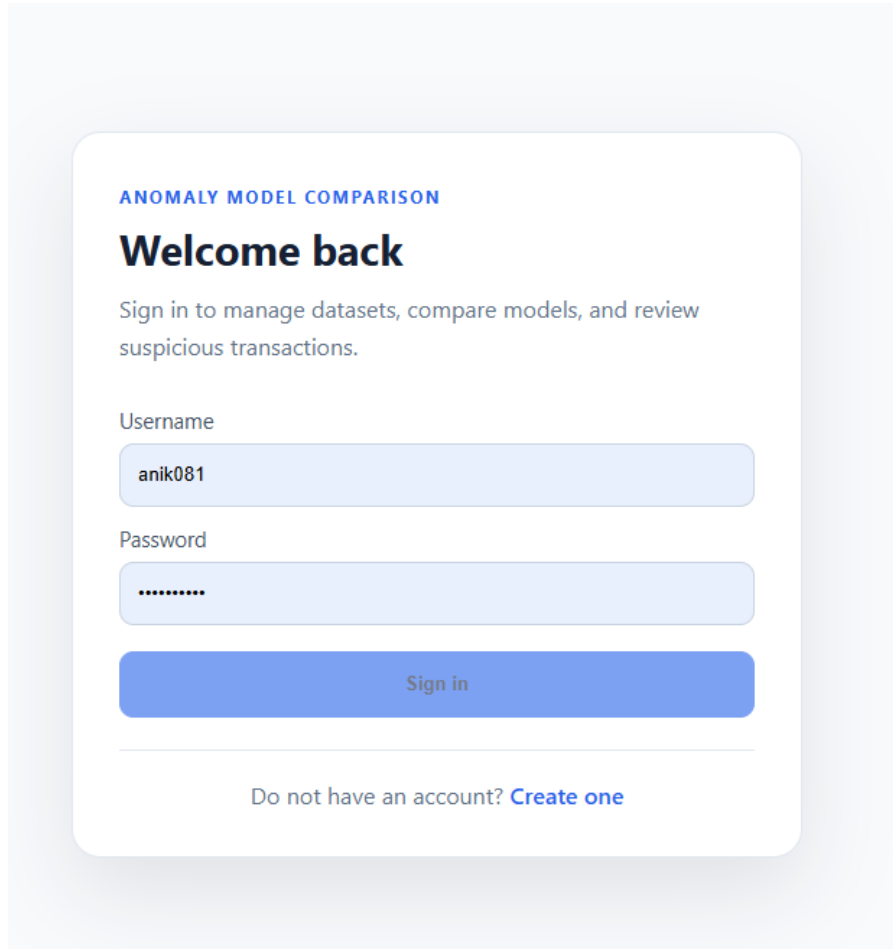
The image shows a login page for a system titled "ANOMALY MODEL COMPARISON". The page has a light blue background. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle, the title "ANOMALY MODEL COMPARISON" is written in blue. Below it, the text "Welcome back" is displayed in a large, bold, black font. Underneath, a subtitle in a smaller black font reads: "Sign in to manage datasets, compare models, and review suspicious transactions." The form includes two input fields: "Username" with the value "anik081" and "Password" with masked characters "*****". Both fields have light blue borders. Below the password field is a blue "Sign in" button. At the bottom of the white rectangle, there is a link that says "Do not have an account? Create one".

Figure 5.1. Login page used to establish an authenticated application session.

Credentials are submitted to the Spring Boot authentication endpoint and a signed JWT is returned after successful verification.

The Angular interceptor attaches the token to protected API requests, while backend role checks restrict administrative operations.

5.9.2 System Settings and Cold Start

Settings controls whether the application operates in supervised or unsupervised mode and defines how accounts with insufficient history are treated.

Anomaly Intelligence...
Model Comparison

Settings

Choose the system-wide learning method and configure how new accounts are handled. The selected System Type controls model training, production scoring, model configuration, and the comparison methodology across the application.

System Type

System Type
Supervised Learning — FraudLabel required

Global learning mode. Allowed values: UNSUPERVISED or SUPERVISED.

Cold Start Settings

New accounts lack enough history for reliable customer and peer features. Cold start can return **NORMAL / ALLOW** until the configured history threshold is reached.

Cold Start Handling
Enabled

Minimum Transaction History
5

Enable or disable cold start handling

Minimum prior transaction count required before ML prediction

Save Settings

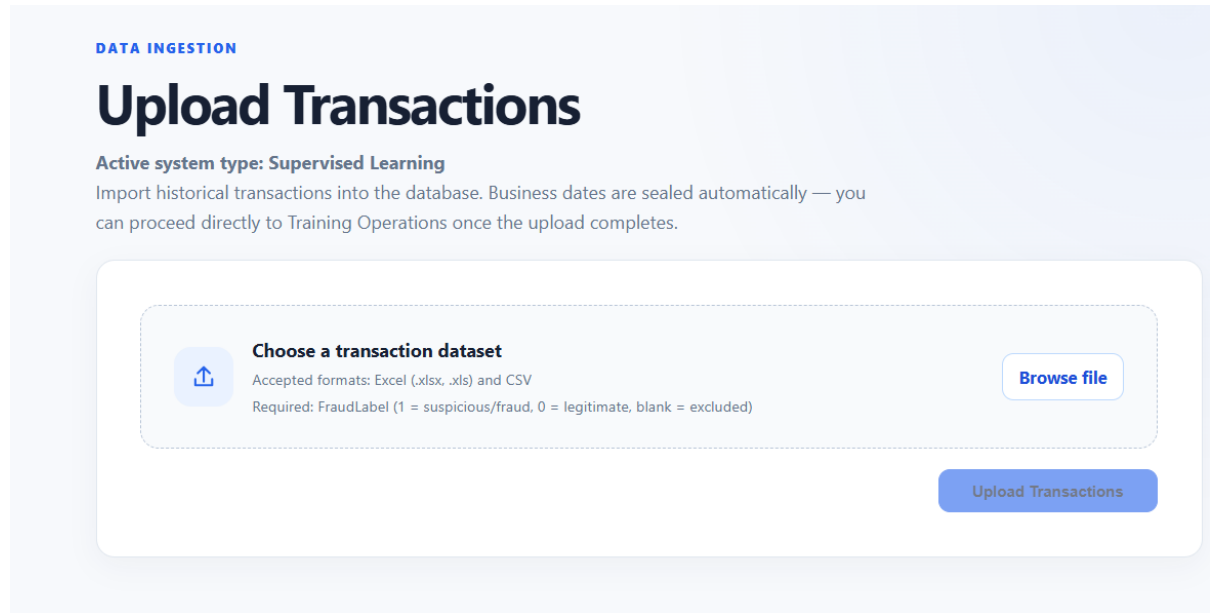
Figure 5.2. System-wide learning mode and cold-start configuration.

Supervised mode requires FraudLabel values and activates XGBoost, Class-Balanced Random Forest, and Extra Trees workflows.

Cold-start handling prevents unstable customer-history features from being treated as reliable before the configured minimum history is reached.

5.9.3 Transaction Dataset Upload

The upload page communicates the active system type and validates whether labelled or unlabeled input is expected before ingestion begins.



DATA INGESTION

Upload Transactions

Active system type: Supervised Learning

Import historical transactions into the database. Business dates are sealed automatically — you can proceed directly to Training Operations once the upload completes.



Choose a transaction dataset

Accepted formats: Excel (.xlsx, .xls) and CSV

Required: FraudLabel (1 = suspicious/fraud, 0 = legitimate, blank = excluded)

[Browse file](#)

[Upload Transactions](#)

Figure 5.3. Mode-aware transaction upload page.

CSV and Excel files are accepted, normalized, validated, and inserted through the backend upload pipeline.

In supervised mode FraudLabel 1 means confirmed suspicious/fraud, FraudLabel 0 means legitimate, and blank labels are excluded from supervised learning.

5.9.4 Training Operations

An operator chooses the historical period and enabled models. The backend creates one immutable feature snapshot so all selected models receive equivalent data.

The screenshot displays a two-step workflow for model training.
Step 1: Seal business date - This step involves selecting a business date. A date picker shows '08/12/2026'. A 'Close day' button is present. A note states: 'Uploaded data is sealed automatically. Use this to close a specific date for live transaction data ingested through the real-time pipeline.'
Step 2: Start training pipeline - This step includes several configuration options:
- **Feature version:** 'AML_FEATURES_V2'
- **From date:** '08/12/2026'
- **To date:** '08/12/2026'
- **Cutoff timestamp:** '08/12/2026 11:59 PM'
- **Choose models for this run:** Three models are listed with checkboxes: 'XGBoost' (checked), 'Random Forest' (checked), and 'Logistic Regression' (checked). A note indicates 'Supervised mode — only labelled rows are used.'
- **FraudLabel is required:** A note explains: 'Rows marked 1 are confirmed suspicious/fraud, rows marked 0 are confirmed legitimate, and blank labels are excluded from supervised training.'
- **What happens after you click:** A list of three steps: 1. A training run is created and locked to the selected date range. 2. Feature rows are exported into an immutable snapshot. 3. 3 selected models train independently on that snapshot. A note below states: 'Risk Policy determines which model scores count toward the fraud decision.'
- A green 'Create & Train' button is at the bottom of the step.
Recent pipeline activity: A table titled 'Training runs' shows a 'Production rebuild' run on '2023-01-01 -- 2026-12-31' with '25,000' rows and a 'CANDIDATE READY' status. The table is paginated, showing '1-4 of 9'.

Figure 5.4. Controlled model-training workflow and recent pipeline activity.

Uploaded business dates are sealed automatically, while live API dates can be closed explicitly before training.

Candidate rows show the number of labelled records actually learned rather than the larger mixed snapshot row count.

5.9.5 Model Tuning

The page exposes only controls consumed by training or comparison, including chronological holdout settings, reproducibility seed, class-imbalance handling, and model-specific hyperparameters for XGBoost, Class-Balanced Random Forest, and Extra Trees.

The screenshot displays the 'Model Tuning' interface, which is part of the 'TRAINING CONTROLS' section. The main heading is 'Model Tuning', with a subtitle 'Tune the active supervised fraud classifiers and their shared comparison protocol. Supervised Learning'. A button labeled 'Reload saved values' is in the top right. A blue banner states: 'Settings apply to the next training run. Changes here do not overwrite existing trained candidates. Production model selection is configured on the Risk Policy page.'

The interface is divided into four main sections:

- Evaluation Protocol** (LAS icon): Controls chronological holdout size, evaluation row cap, minimum partition data, and the shared reproducibility seed used across all training and comparison runs.
 - Minimum partition rows**: Input field with value 200. Description: 'Skip comparison partitions with fewer rows than this limit.' Allowed range: 10-100000.
 - Chronological holdout fraction**: Input field with value 0.1. Description: 'Newest fraction of each partition reserved for evaluation.' Allowed range: 0.1-0.4.
 - Evaluation row cap**: Input field with value 20000. Description: 'Maximum evaluation rows scored per model and partition.' Allowed range: 100-1000000.
 - Random seed**: Input field with value 42. Description: 'Shared reproducibility seed for training and comparison.' Allowed range: 0-2147483647.
- XGBoost** (XGB icon): Gradient boosted decision trees optimized for nonlinear fraud classification on labelled transactions.
 - Training enabled**: Dropdown menu set to 'Enabled'. Description: 'Generate new XGBoost fraud classifier candidates.'
 - Boosting trees**: Input field with value 300. Description: 'Number of sequential boosted trees.' Allowed range: 10-2000.
 - Maximum tree depth**: Input field with value 6. Description: 'Maximum interaction depth of each tree.' Allowed range: 2-16.
 - Learning rate**: Input field with value 0.05. Description: 'Contribution of each new tree.' Allowed range: 0.001-0.5.
 - Row sample fraction**: Input field with value 0.8. Description: 'Fraction of labelled rows sampled for each tree.' Allowed range: 0.5-1.
 - Feature sample fraction**: Input field with value 0.8. Description: 'Fraction of features sampled for each tree.' Allowed range: 0.5-1.
- Random Forest** (RF icon): Class-balanced tree ensemble providing a robust supervised benchmark.
 - Training enabled**: Dropdown menu set to 'Enabled'. Description: 'Generate new Random Forest fraud classifier candidates.'
 - Number of trees**: Input field with value 300. Description: 'Number of independently trained trees.' Allowed range: 10-2000.
 - Maximum tree depth**: Input field with value 12. Description: 'Limits tree complexity and memorization.' Allowed range: 3-50.
 - Minimum leaf rows**: Input field with value 2. Description: 'Minimum labelled rows in each terminal leaf.' Allowed range: 1-100.
- Logistic Regression** (LR icon): Regularized, interpretable probability baseline for labelled fraud detection.
 - Training enabled**: Dropdown menu set to 'Enabled'. Description: 'Generate new Logistic Regression fraud classifier candidates.'
 - Inverse regularization strength**: Input field with value 1.0. Description: 'Lower values apply stronger regularization.' Allowed range: 0.001-100.

At the bottom, a section titled 'Ready to apply these settings?' states: 'The next applicable training or research run will read these values from app.config.' A blue button labeled 'Save Model Tuning' is in the bottom right.

Figure 5.5. Supervised model parameters and shared evaluation protocol.

Changes apply to subsequent runs and do not silently rewrite already registered model artifacts.

Training enable switches determine which models are available in Training Operations.

5.9.6 Risk Decision Policy

The policy combines customer behavior, peer behavior, the selected ML ensemble, and deterministic AML rules into a normalized score from 0 to 100.

RULE CONFIGURATION

Risk Decision Policy

Choose the weight of the rules, split the 100% score, and train them manually when needed.

Active policy
AML_RISK_POLICY_20260812094330028
Updated Aug 12, 2026, 9:43:30 PM

FINAL SCORE
Top-level layers plus a nested ML ensemble

Medium starts at 65% and creates a case automatically. Use a component's remaining-budget button to make only that component complete the 100% total.

1 Allocate Rule Weights

Edit layers independently. Saving is available when the total reaches exactly 100%.

Component	Weight	Remaining Budget
Customer behaviour	24%	24%
Peer behaviour	9%	9%
ML ensemble	65%	65%
AML rules	2%	2%

2 Define risk and rollout thresholds

Medium is the suspicious boundary and automatically creates a case.

Threshold	Value
LOW starts	40% (0.4)
MEDIUM / create case	65% (0.65)
HIGH starts	80% (0.8)

Normal **Low** **Medium** **High**

Saving creates a new policy version
Run shadow validation and promotion from Model Governance before this strategy controls production routing.

Save policy

Figure 5.6. Layered risk weighting and operational decision thresholds.

Each component can consume its remaining percentage without unexpectedly changing unrelated controls during manual adjustment.

The medium threshold creates a review case, while the high threshold can trigger automatic STR preparation according to policy.

5.9.7 Unsupervised Current-Model Comparison

This comparison is used when confirmed fraud labels are unavailable. It reports label-free distribution diagnostics rather than claiming fraud accuracy.

Current models			
Analysis			
Metric	ISOLATION FOREST UNSUPERVISED	AUTOENCODER UNSUPERVISED	LOCAL OUTLIER FACTOR UNSUPERVISED
Learned rows	50,000	50,000	50,000
EM-AUC COMPARABLE	67.2 / 100	85.6 / 100	89.2 / 100
Score skewness COMPARABLE	77.5 / 100	100.0 / 100	100.0 / 100
Anomaly rate	1.00%	2.50%	0.82%
Average score NATIVE SCALE	-0.057	-0.257	-0.490
Score P95 NATIVE SCALE	-0.021	-0.137	-0.266
Score P99 NATIVE SCALE	0.000	0.401	-0.033
Alert count	500	1,250	412
Validation rows	50,000	50,000	50,000
Training duration	1,309 ms	3,194 ms	5,595 ms
Registered at	Aug 17, 2024, 9:18:17 AM	Aug 17, 2024, 9:18:17 AM	Aug 17, 2024, 9:18:17 AM
Feature version	PRE_FRAUDRISK_V2	PRE_FRAUDRISK_V2	PRE_FRAUDRISK_V2

Figure 5.7. Current Isolation Forest, Autoencoder, and Local Outlier Factor diagnostics.

EM-AUC and score-skewness values help compare score separation across detector types.

Anomaly rate, alert count, training duration, and learned rows expose operational cost and alert-volume implications.

5.9.8 Unsupervised Model Agreement

The agreement study determines whether models flag the same records or contribute complementary anomaly evidence.



Figure 5.8. Pairwise overlap and consensus among unsupervised detectors.

Jaccard overlap measures shared flagged sets without requiring class labels.

Exactly-one, exactly-two, and unanimous counts help configure ensemble voting and understand alert concentration.

5.9.9 Unsupervised Growth Analysis

The oldest 10%, 25%, 50%, and 100% partitions are evaluated to show whether detector behaviour stabilizes as more history becomes available.

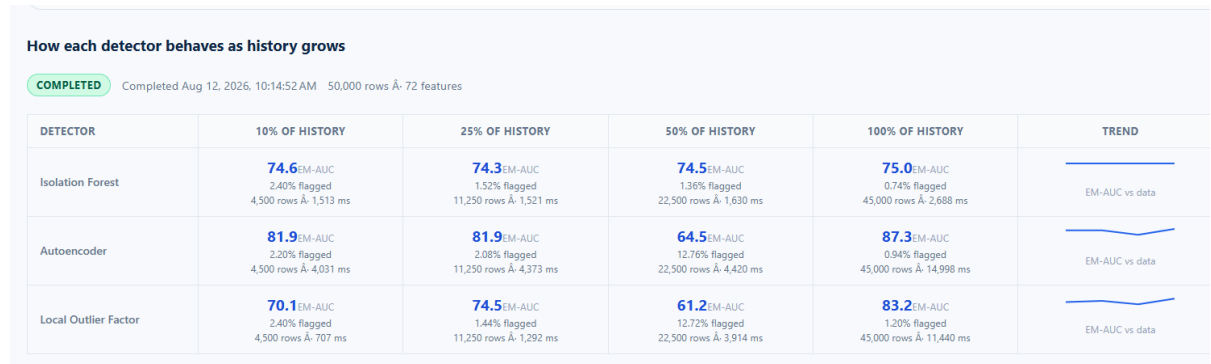


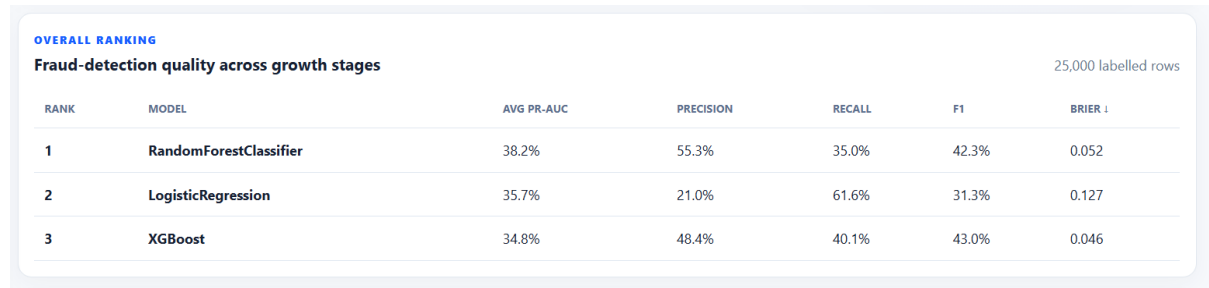
Figure 5.9. Detector behaviour as chronological history grows.

Stored growth studies prevent expensive multi-model analysis from rerunning whenever the page is opened.

Trend lines make changes in separation, anomaly rate, and training cost visible to non-technical reviewers.

5.9.10 Supervised Model Ranking

The ranking compares PR-AUC, precision, recall, F1, calibration, and throughput using chronological evaluation rather than random future-data leakage.



OVERALL RANKING						
Fraud-detection quality across growth stages						
25,000 labelled rows						
RANK	MODEL	AVG PR-AUC	PRECISION	RECALL	F1	BRIER 1
1	RandomForestClassifier	38.2%	55.3%	35.0%	42.3%	0.052
2	LogisticRegression	35.7%	21.0%	61.6%	31.3%	0.127
3	XGBoost	34.8%	48.4%	40.1%	43.0%	0.046

Figure 5.10. Labelled fraud-classification ranking across growth stages.

The latest implementation also reports accuracy, balanced accuracy, PR-AUC lift, calibrated decision thresholds, and a confusion matrix.

Results are cached per immutable snapshot, so reopening the page reads stored evidence instead of retraining twelve model-partition combinations.

5.9.11 Supervised Growth Matrix

The matrix reveals whether model quality improves, declines, or remains stable as labelled history expands.

GROWTH MATRIX				
Performance at every labelled partition				
MODEL	10%	25%	50%	100%
XGBoost	28.7% PR-AUC 53.3% precision · 30.8% recall 39.0% F1 · 159,785 rows/s	33.6% PR-AUC 50.0% precision · 36.5% recall 42.2% F1 · 329,615 rows/s	36.7% PR-AUC 50.5% precision · 42.9% recall 46.4% F1 · 616,189 rows/s	40.4% PR-AUC 39.8% precision · 50.2% recall 44.4% F1 · 538,370 rows/s
RandomForestClassifier	37.8% PR-AUC 53.8% precision · 26.9% recall 35.9% F1 · 8,685 rows/s	35.2% PR-AUC 57.1% precision · 30.8% recall 40.0% F1 · 18,170 rows/s	37.7% PR-AUC 58.7% precision · 37.0% recall 45.4% F1 · 24,571 rows/s	42.2% PR-AUC 51.4% precision · 45.1% recall 48.1% F1 · 62,029 rows/s
LogisticRegression	35.6% PR-AUC 22.5% precision · 69.2% recall 34.0% F1 · 1,681,237 rows/s	36.2% PR-AUC 17.1% precision · 57.7% recall 26.4% F1 · 849,762 rows/s	33.9% PR-AUC 21.1% precision · 60.5% recall 31.2% F1 · 1,814,487 rows/s	37.0% PR-AUC 23.3% precision · 59.1% recall 33.5% F1 · 5,243,289 rows/s

Figure 5.11. Supervised performance at every labelled-data partition.

PR-AUC is emphasized because fraud is rare; plain accuracy alone can appear high even when minority-class detection is weak.

On the 25,000-row demonstration snapshot, Random Forest achieved approximately 95.3% accuracy, 70.5% balanced accuracy, and 8.9 times PR-AUC lift over the fraud prevalence baseline.

5.9.12 Manual Case Creation

Investigators can search transaction or account identifiers, inspect the selected record, set priority and assignment, and create a case without waiting for automatic scoring.

On-demand investigation

Create Manual Case

Search existing transactions and open a review case without waiting for anomaly voting.

Back to Case Management

Step 1

25 results

Find a transaction

Search by transaction ID or account ID

Search

SUP-00025000

371.29

AC00468 · TRANSFER · ATM
Dec 31, 2025, 11:36:44 PM · London

SUP-00024999

759.06

AC00394 · DEBIT · ATM
Dec 31, 2025, 10:26:35 PM · Rajshahi

SUP-00024998

82.37

AC00246 · DEBIT · ONLINE
Dec 31, 2025, 10:24:18 PM · Rangpur

SUP-00024997

255.46

AC00109 · CREDIT · ATM
Dec 31, 2025, 3:00:28 PM · Rangpur

SUP-00024996

50.20

Step 2

Review and create

Selected transaction

SUP-00025000

Account	Amount
AC00468	371.29
Date	Balance
Dec 31, 2025, 11:36:44 PM	118,141.21
Location	Source
London	BULK_UPLOAD

Case title

Manual review for transaction SUP-00025000

Priority

Assign to

Medium

Optional analyst username

Create Manual Case

Figure 5.12. On-demand case creation from an existing transaction.

The workflow supports intelligence-led review and external referrals that are not initiated by the ML ensemble.

The selected transaction remains the auditable source record for the manually opened case.

5.9.13 Case Management

Automatically generated and manual cases are listed in a paginated review queue with status, priority, transaction, account, assignment, and decision evidence.

The screenshot displays the 'Case Management' interface within an 'Investigation workspace'. At the top, a header bar contains the title 'Case Management' and a 'Create Manual Case' button. Below the header, a summary row shows four metrics: 'All cases' (1), 'Active review' (1), 'STR generated' (0), and 'False positive' (0). The main content area is divided into two panels. The left panel, titled 'Review queue', includes a 'Refresh' button and a 'Generated cases' section with a search bar and a dropdown menu for 'Active cases'. A single case is listed: 'CASE-1786529744682' with a 'MEDIUM' priority and an 'OPEN' status. The right panel displays the details for the selected case, 'Automatic anomaly case for transaction TXN-1786529728924'. It includes a 'CASE-1786529744682 · opened Aug 12, 2026, 4:15:44 PM' header and a table of attributes: Transaction (TXN-1786529728924), Account (AC00455), Priority (MEDIUM), and Assigned to (Unassigned). Below this, the 'Decision evidence' section shows a 'Model prediction snapshot' with a 'MEDIUM' priority. The snapshot includes three data points: 'Anomaly votes' (2), 'Isolation Forest score' (0.02371), and 'Model version' (Legacy model).

Figure 5.13. Investigation queue, case evidence, and review actions.

Investigators can mark a case as a false positive or generate an STR when evidence supports escalation.

Human-readable reason descriptions explain why the transaction was considered unusual instead of exposing only technical reason codes.

TESTING AND RESULTS

6.1 Testing Strategy

Testing combined automated unit and integration tests with manual end-to-end acceptance checks. Java tests cover feature context, customer and peer scoring, rules, layered risk arithmetic, training export, model registry, validation, deployment, case management, authentication, and comparison services. Python tests cover growth analysis, persisted-feature alignment, artifact bundles, route compatibility, finite-value checks, and model artifact reuse. Angular production builds verify template and TypeScript integration.

6.2 Dataset Sources and Evidence Boundaries

Two evidence tracks are intentionally separated. The unsupervised track uses an unlabeled 50,000-row development snapshot and therefore reports label-free diagnostics only. The main supervised benchmark uses the open Fraud Detection Handbook simulated-data-transformed source created by Le Borgne, Siblini, Lebichot, and Bontempi. The project selected complete customer histories and preserved source chronology, amounts, terminal identity, and TX_FRAUD labels; TERMINAL_ID is mapped to Location so the application can derive novelty without using the label as a feature.

Evidence set	Rows / period	Purpose and boundary
Unlabeled development snapshot	50,000 rows	Unsupervised EM-AUC, score shape, anomaly rate, runtime. No fraud-accuracy claim.
Fraud Detection Handbook adapted subset	264,500 rows; 749 customers; 183 days; 1 Apr-30 Sep 2018	Main reproducible supervised benchmark; 2,340 fraud rows (0.8847%).
Application integration snapshot	25,000 labelled rows	End-to-end UI, fusion, growth, and four-layer policy replay; synthetic demonstration evidence.

Leakage boundary. Fraud labels and fraud-scenario identifiers are never model inputs. Training uses the oldest 70%, threshold calibration uses the next 10%, and final reporting uses the newest untouched 20% for the reproducible benchmark.

6.3 Current Unsupervised Diagnostic Result

Current models Analysis			
Metric	ISOLATION FOREST UNSUPERVISED	AUTOENCODER UNSUPERVISED	LOCAL OUTLIER FACTOR UNSUPERVISED
Learned rows	50,000	50,000	50,000
EM-AUC COMPARABLE	67.2 / 100	85.6 / 100	89.2 / 100
Score skewness COMPARABLE	77.5 / 100	100.0 / 100	100.0 / 100
Anomaly rate	1.00%	2.50%	0.82%
Average score NATIVE SCALE	-0.057	-0.257	-0.490
Score P95 NATIVE SCALE	-0.021	-0.117	-0.255
Score P99 NATIVE SCALE	0.000	0.401	-0.831
Alert count	500	1,250	412
Validation rows	50,000	50,000	50,000
Training duration	1,309 ms	3,194 ms	5,595 ms
Registered at	Aug 12, 2019, 9:18:17 AM	Aug 12, 2019, 9:18:17 AM	Aug 12, 2019, 9:18:17 AM
Feature version	APL_FEATURES_V2	APL_FEATURES_V2	APL_FEATURES_V2

Figure 8. Current label-free diagnostics on the 50,000-row unsupervised snapshot

Detector	EM-AUC	Score skewness	Anomaly rate	Training time
Isolation Forest	67.2 / 100	77.5 / 100	1.00%	1,309 ms
Autoencoder	85.6 / 100	100.0 / 100	2.50%	3,194 ms
Behavioral Cluster Outlier	89.2 / 100	100.0 / 100	0.82%	5,595 ms

Behavioral Cluster Outlier currently has the strongest EM-AUC and the lowest anomaly rate, while Isolation Forest is fastest. Autoencoder also provides strong score separation. These measurements compare anomaly-score structure; they do not establish precision, recall, or confirmed fraud detection.

6.4 Current Reproducible Supervised Benchmark

On the 264,500-row Fraud Detection Handbook subset, Extra Trees produced the strongest balanced operating point on the newest untouched 20%. The threshold was chosen only on the calibration period to maximize recall while maintaining at least 70% calibration precision.

Metric	Extra Trees result	Interpretation
PR-AUC	73.3%	Strong ranking quality under 0.8847% fraud prevalence
Precision	74.8%	323 of 432 fraud alerts were true fraud
Recall	71.8%	323 of 450 fraud transactions were captured
F1	73.2%	Balanced precision-recall operating point
Confusion matrix	TP 323; FP 109; FN 127; TN 52,341	Untouched newest 52,900 transactions

Random Forest reached 74.9% precision and 66.4% recall at its precision-constrained threshold. XGBoost reached 81.9% precision but only 48.2% recall. The result demonstrates why the project compares multiple algorithms and selects an operating point instead of manipulating a fixed 90% precision target.

6.5 Measured Fusion and Full-Policy Replay

INDIVIDUAL MODELS						
Standalone model performance						
RANK	MODEL	PR-AUC	PRECISION	RECALL	F1	BALAN
1	XGBoost	52.3%	58.4%	65.2%	61.6%	81.5%
2	RandomForestClassifier	52.0%	57.5%	65.7%	61.2%	81.6%
3	ExtraTreesClassifier	48.8%	55.8%	66.9%	60.7%	82.1%

FUSION STRATEGIES						
Decisions built from the same three base classifiers						
DECISION STRATEGY	PR-AUC	PRECISION	RECALL	F1	BALANCED	
Weighted probability ensemble	50.8%	58.7%	65.9%	62.0%	81.8%	
Leakage-safe temporal stacking	51.8%	59.5%	64.0%	61.5%	80.9%	
Majority vote (2 of 3)	43.3%	57.5%	66.3%	61.5%	81.9%	
Any model (1 of 3)	43.3%	54.5%	70.3%	61.4%	83.7%	
Unanimous vote (3 of 3)	43.3%	60.0%	61.2%	60.5%	79.6%	

Figure 9. Measured standalone and fusion results on the 25,000-row integration snapshot

On the same 25,000 held-out integration rows, fusion did not materially outperform the strongest standalone classifier. The weighted probability ensemble produced 58.7% precision, 65.9% recall, and 62.0% F1. Temporal stacking remains a research comparison but has been removed from the production-model selection UI; production uses weighted base classifiers only.

Policy PR-AUC

48.9%

Continuous final score

Case accuracy

95.7%

Read with class-balanced metrics

Balanced accuracy

74.1%

Equal importance to fraud and legitimate rows

Case precision

54.6%

-1.4 pts vs ML-only

Case recall

50.2%

119 captured · 118 missed

Case F1

52.3%

-6.9 pts vs ML-only

Case volume

43.6

cases per 1,000 transactions

STR precision

66.3%

92 HIGH decisions

Frozen policy used

Customer behaviour25.0%

Peer behaviour10.0%

ML ensemble60.0%

AML rules5.0%

PRODUCTION ML COMPOSITION

Class-Balanced Random Forest33.3%

XGBoost33.3%

Extra Trees33.3%

Case decision (MEDIUM/HIGH)

74.1% balanced

No case

Create case

Actual legitimate

TN 4,664

93.3%

FP 99

2.0%

Actual fraud

FN 118

2.4%

TP 119

2.4%

STR decision (HIGH)

62.5% balanced

No STR

Generate STR

Actual legitimate

TN 4,732

94.6%

FP 31

0.6%

Actual fraud

FN 176

3.5%

TP 61

1.2%

Figure 10. Measured four-layer risk-policy replay on the same held-out integration rows

Measured system decision	Precision	Recall	F1 / balanced	Operational volume
Create case (MEDIUM/HIGH)	54.6%	50.2%	52.3% F1; 74.1% balanced accuracy	43.6 cases per 1,000

Generate STR (HIGH)	66.3%	25.7%	62.5% balanced accuracy	92 HIGH decisions
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The frozen policy used 25% customer behaviour, 10% peer behaviour, 60% ML ensemble, and 5% AML rules. Its lower case-level result shows that adding layers does not automatically improve classification. Customer, peer, and rule layers serve explainability and AML controls, but their weights and thresholds still require calibration on a separate validation period. This limitation is reported rather than hidden.

6.6 Explainable Transaction Scenario

In a tested scenario, the selected ML detectors flagged an anomaly while customer, peer, and rules supplied separate reasons such as amount deviation, peer-frequency deviation, and turnover thresholds. The UI shows model diagnostics, normalized component scores, reasons, the final 0-100 score, and the policy version. MEDIUM creates a review case; HIGH also prepares a draft STR. This keeps a suspicious decision explainable and reviewable rather than claiming autonomous confirmation of fraud.

6.7 Automated Validation and Scalability

Angular and Spring Boot production builds pass; the current Angular bundle reports only existing size-budget warnings.

The 264,500-row model pipeline avoids the earlier 19.8 GiB one-hot allocation by using compact bounded feature representations.

Training snapshots are immutable and checksummed; model artifacts are versioned and do not become active automatically.

Online scoring reuses loaded artifacts and bounded point-in-time features; case and STR work occurs after the transaction decision path.

6.8 Scientific Interpretation Limits

The presentation-only cache is never treated as measured evidence. Unsupervised EM-AUC is not fraud accuracy. The 25,000-row integration snapshot is synthetic demonstration data. The 264,500-row Fraud Detection Handbook benchmark is reproducible simulator evidence, not a claim about a bank's live population. A production claim requires institution-specific labels, temporal drift monitoring, calibration,

investigator-capacity metrics such as precision at k, fairness checks, and independent validation.

CHAPTER 7

PROBLEMS AND SOLUTIONS

7.1 Problem-to-Solution Summary

Table 8. Problem-to-solution matrix

Problem	Implemented solution	Result
No reliable fraud labels	Global unsupervised mode with Isolation Forest, Autoencoder, and Behavioral Cluster Outlier; label-free metrics	Useful anomaly research without false accuracy claims
Some clients have reviewed labels	Separate supervised mode with XGBoost, Class-Balanced Random Forest, Extra Trees, and supervised metrics	One platform supports both scientifically valid workflows
Model behavior changes with data volume	Oldest-first 10%, 25%, 50%, 100% growth study with chronological holdout	Stability and scalability become visible before selection
Raw scores are incompatible	Normalize every model and component to 0-1 before weighted aggregation	All configured contributions total exactly 100%
ML alone misses AML obligations	Separate deterministic AML rules with hard override support	Mandatory controls remain transparent and auditable
One customer differs from population	Customer behavior scorer with amount, velocity, novelty, time-gap, and confidence	Personal deviations contribute independently
Peer comparison unavailable for sparse groups	Occupation-age group, parent segment, then global fallback with confidence reduction	Every transaction can receive a bounded peer result
Cold start creates unstable profiles	Minimum history gate and confidence multiplier; suspicious rows excluded from trusted learning	Immature or contaminated profiles have less authority
Training and prediction feature drift	Persisted versioned feature vectors and exact feature-column artifact	Training and scoring use the same schema

Problem	Implemented solution	Result
Millions of rows cannot be sent as JSON	Keyset export, Parquet parts, checksums, capped evaluation, selected-model training	Memory and network usage remain bounded
Transaction checks were slow	Cached model artifacts, persisted features, bounded history, asynchronous post-commit case work	Decision path avoids retraining and case I/O
Model votes conflicted with final score	Display votes as diagnostics and calculate final score from explicit policy weights	Users can explain why 3/3 anomalies may still be below threshold
Weight sliders changed unrelated values	Normalize within each component and model section; enforce totals server-side	Policy editing is predictable and cannot exceed 100%
Cases became expensive to list	Server-side pagination, filtering, count summaries, and reduced N+1 loading	Case management scales beyond small demonstrations
Alerts did not complete the AML workflow	Automatic cases, notes, false-positive outcome, manual cases, and draft STR XML	Suspicious decisions become reviewable operational records
Administrative APIs were exposed	JWT authentication, BCrypt, role authorization, deny-by-default security	Sensitive operations require authorized identities
Training could silently change production	Immutable model registry, validation, version locking, controlled promotion, rollback	Model lifecycle is auditable and reversible

7.2 How Scalability Is Solved

Scalability is addressed by separating the online decision path from offline training. Online scoring reads bounded point-in-time state and already loaded artifacts. Offline training uses closed windows and immutable Parquet snapshots. The UI sends a training-run identifier rather than millions of feature rows. Evaluation rows are capped, large algorithms can use bounded samples, and candidate creation is manual and asynchronous. SQL Server remains the governance store rather than the numerical training engine.

7.3 Kafka Integration Roadmap

Kafka is not required for the current prototype, but it is the next transport layer for millions of daily events. A SQL outbox should publish committed transaction events to account-keyed topics. Independent consumers can update customer, peer, and velocity state; write feature events; request predictions; persist model results; and create cases. Account-keyed partitioning preserves ordering for one customer while consumer groups

provide horizontal scaling, replay, and back-pressure isolation. Kafka transports events; it does not replace SQL governance or train models by itself.

7.4 Why the Final Design Is More Flexible

The client decides the learning methodology once at system level, chooses participating production models based on comparison evidence, assigns model and layer weights, and manually starts training when data is ready. The same business layers continue to operate in both modes. This avoids forcing a bank with no labels into supervised metrics and avoids preventing a bank with reviewed labels from using probability-based classifiers.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

The project delivers a complete prototype for transaction ingestion, feature persistence, dual-mode machine learning, model growth comparison, explainable layered risk scoring, governed training, authentication, and AML case management. Its most important design decision is methodological honesty: anomalies are treated as suspicious evidence, not confirmed fraud, unless reviewed labels justify supervised evaluation.

The system solves the central project problems by using point-in-time features, immutable chronological datasets, mode-specific models and metrics, normalized risk weights, transparent customer and peer scorers, deterministic rules, auditable model versions, and a reviewer workflow. The Angular interface makes these decisions visible rather than hiding them behind one prediction label.

8.2 Current Limitations

- The 50,000-row result is a development-scale experiment, not a production bank benchmark.
- Unsupervised results cannot provide true precision, recall, F1, or confirmed false-positive rate.
- The generated supervised dataset is synthetic and must not replace reviewed institutional labels for real deployment.
- Kafka, distributed feature state, centralized observability, and infrastructure deployment are architectural roadmaps rather than completed production services.
- Draft STR XML requires validation against the exact schema and submission process of the relevant regulator.
- Model calibration, fairness by segment, privacy controls, penetration testing, and disaster recovery require further work.

8.3 Future Work

- Integrate Kafka through a transactional SQL outbox and account-keyed consumer groups.
- Add online feature stores or bounded materialized state for customer and peer statistics.
- Collect reviewer labels and add calibration curves, threshold optimization, recall at fixed false-positive rates, and drift by business period.
- Run reproducible load tests with millions of rows and concurrent transaction scoring, reporting p50, p95, and p99 latency.
- Introduce approved clustering experiments using MiniBatchKMeans or BIRCH and display PCA/UMAP projections with Silhouette, Davies-Bouldin, and Calinski-Harabasz as geometric diagnostics.
- Add containerization, centralized logs and metrics, secrets management, backup, and automated deployment pipelines.
- Validate STR XML and workflow against the applicable jurisdiction and add digital approval controls.

8.4 Final Remarks

The completed platform demonstrates that a strong fraud and AML project is not only a model-selection exercise. It is a data-integrity, scalability, explainability, governance, and human-review problem. By making each of those concerns explicit, the project provides a credible foundation for further academic study and future production engineering.

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APPENDIX A: MAIN USER WORKFLOW

1. Log in with an authorized account.
2. Open Settings and choose supervised or unsupervised learning.
3. Upload CSV/Excel data and inspect the batch result.
4. Close the relevant business dates in Training Operations.
5. Create a training run, generate the immutable dataset, and wait for DATASET_READY.
6. Choose the desired active-mode models and start training.
7. Open the corresponding comparison page and review quality, stability, anomaly/classification metrics, and runtime.
8. Open Risk Policy, select production models, and configure normalized layer/model weights and thresholds.
9. Use Transaction Check or the transaction API to score new activity.
10. Review automatically generated cases; mark false positives or generate/download STR XML as appropriate.

APPENDIX B: SELECTED API ENDPOINTS

Table B1. Selected API endpoints

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate and obtain JWT
POST	/api/v1/uploads/transactions	Upload CSV or Excel transactions
POST	/api/v1/transactions	Persist and score one transaction
POST	/api/v1/aml/training-runs/pipeline/start	Create governed training run
POST	/api/v1/aml/training-runs/{id}/dataset	Generate immutable feature dataset
POST	/api/v1/aml/growth-analysis/training-runs/{id}	Run chronological growth analysis
GET/PUT	/api/v1/anomaly-model-comparisons/risk-policy	Read or save production risk policy
GET/PUT	/api/v1/anomaly-model-comparisons/model-tuning	Read or save active-mode tuning
GET	/api/cases	Paginated case search
POST	/api/cases/{id}/false-positive	Record false-positive outcome
POST	/api/cases/{id}/str-xml	Generate draft STR XML

APPENDIX C: CONFIGURATION PRINCIPLES

- Top-level risk-layer weights must total 100%.
- Enabled model weights inside the ML ensemble must total 100%.
- Supervised mode requires reviewed labels; unresolved labels are excluded from training.
- Unsupervised metrics must never be described as confirmed fraud accuracy.
- Training uses closed business dates and a fixed cutoff.
- No training run automatically activates or promotes a model.
- MEDIUM creates a case; HIGH creates a case and draft STR XML under the current policy.
- The backend is the security authority even when UI controls are hidden or disabled.